

C1

Where is AI worth building?

~49 min · the value gap · the AI factory · data readiness · what a use case buys

BY THE END OF THIS CAPSULE YOU WILL BE ABLE TO

Decide which AI use cases are worth doing — and defend the ranking on what the use case buys, whether your data can feed it, and whether your operating model can collect it.

88% of organisations use AI. What share can find any of it in their EBIT?

A · About 80%

B · About 60%

C · About 39%

D · About 15%

EBIT = earnings before interest and taxes — the profit line. “Any impact at all”, not a big one.

Adoption is finished. Value is not.

Four numbers from the same survey. The distance between the first and the last is this capsule.

The adoption story is over. 88% use AI in at least one business function, up from 78%; 67% in more than one; 50% across three or more. Generative AI specifically: 79%, up from 65%.

So: if your strategy's achievement is “we are using AI”, you are describing table stakes.

The value story has barely started. Only 33% have begun scaling enterprise-wide. Only 39% report any EBIT impact, most of it below 5%. About 6% clear 5% of EBIT.

1 • PLUMBING

88% using → 33% scaling. The pilot ran on a hand-assembled extract; nobody owns the daily path.

2 • SELECTION

33% scaling → 39% any EBIT. It works perfectly, and automates something that was never expensive.

3 • MEASUREMENT

39% → 6% material. No baseline was captured, so finance cannot book what it cannot attribute.

The survey's strongest discriminator is **PROCESS REDESIGN**: 55% of high performers fundamentally redesigned their workflows, against 20% of everyone else (Pre-read 2). What that does and does not prove is four slides away.

Takeaway Three leaks, three different fixes — and firms answer all three by buying another tool.

MIT reported that 95% of AI pilots return nothing. All four statements below are TRUE. Which one most undermines the headline?

A · The sample was small — 52 interviews, 153 responses

B · It was not peer-reviewed

C · About 80% of that sample never piloted a custom system at all

D · Its authors commercialise the infrastructure it recommends

All four are true. Only one changes the number.

Reading the numbers critically

Three headline claims about the same question, published months apart. They do not contradict.

Claim	Who was asked	What counted as success	Denominator
MIT NANDA (Jul 2025) “95% get zero return”	52 interviews + 153 survey responses, recruited at conferences	Sustained, marked P&L impact within ~6 months	All firms — ~80% never piloted at all
McKinsey (Nov 2025) 39% see any EBIT impact	1,993 respondents, 105 nations, GDP- weighted	Any measurable impact on EBIT	All firms surveyed
Wharton (Oct 2025) 74% report positive ROI	~800 US enterprise decision-makers	Positive ROI, as judged by the buyer	All respondents
The reconciliation	Three different populations	Belief · attribution · a six-month P&L test	Restated on firms that piloted, MIT ≈ 1 in 4 succeeded

Worked: of 100 firms, 80 never piloted, 20 did, 5 succeeded. $5 \div 100 = \text{“95% get zero return”}$. $5 \div 20 = \text{“three in four pilots fail”}$. Same five firms — only the denominator moved.

Takeaway Three questions, every time: who was asked, what threshold, what denominator.

Process redesign: 55% vs 20%

55% of high performers redesigned their workflows; 20% of everyone else did. Three explanations fit that gap.

Three rival explanations	What it says is happening	The lever it hands you
1 • Redesign → performance	Bolting AI onto an unchanged process leaves the handoffs, approvals and batch cadences that actually set cycle time untouched	Redesign the workflow before you automate a step inside it
2 • Capability → both	Firms that <i>can</i> redesign a workflow already have cross-functional authority, documented process and a tolerance for disruption — which predict success at any large programme	Build that capability. Redesigning one workflow is how you build it
3 • Success → redesign	Early wins buy the political capital and the budget to attempt something structural — so the arrow may run backwards	Sequence deliberately: win something small, spend the credibility on the structural change
Not an explanation — the test	Longitudinal data on the same firms, or a natural experiment, holding prior capability constant	Neither exists. Hold the causal claim loosely and the action firmly

Be suspicious of anyone who presents 55-against-20 as a proven lever — including this slide. A cross-sectional survey cannot supply causation (Pre-read 2, Q3).

Takeaway Do not chase the causation. Work all three levers — capability, sequencing, redesign.

Business model vs operating model

The distinction that carries the whole week — and the half that strategy teaching usually skips.

Business model — the promise and the payment. What you sell, to whom, at what price, and why they choose you. Almost all strategy teaching lives here.

The trap: two firms can hold the same business model and not be in the same business at all.

Operating model — how the work actually gets done. People, processes, systems, and the way capacity is added when demand grows. It is the unglamorous half, and it is the half that sets your ceiling. Ant and a retail bank make the same promise — “lend money to people who need it”. Here is how each one keeps it.

ANT • adds capacity by deploying

The ten-thousandth credit decision costs about what the tenth did. Growth is a deployment question.

A RETAIL BANK • adds capacity by hiring

Ten times the loans needs ten times the underwriters — and their supervisors, and their training.

Iansiti & Lakhani, “Competing in the Age of AI”, HBR Jan–Feb 2020 — summarised in Pre-read 1, which is the assigned reading; the article itself is paywalled.

Takeaway AI changes the operating model. The operating model is what limits the business model.

The AI factory — four components

A factory turns a raw input into a manufactured output on a repeatable line. Here the input is data and the output is a decision. Four parts, and they hold each other up.

1 • DATA PIPELINE

A model can only be as timely and correct as the feature it is handed.

Turns data from an asset someone owns into an input the line consumes every second.

2 • ALGORITHM DEVELOPMENT

The pipeline moves data; something must convert it into a judgement.

For most problems, now RENTED by API call — the single biggest change since 2020.

3 • EXPERIMENTATION PLATFORM

Without a controlled comparison you cannot tell improvement from drift or luck.

Converts opinion into evidence, at a rate the organisation can afford.

4 • SOFTWARE INFRASTRUCTURE

Stops the factory being one clever team's private tool.

Buys SCOPE — payments data behind an interface serves lending, insurance and wealth at once.

This is Week 3's eval set and Week 4's harness, one level up: you cannot improve what you are not measuring.

Takeaway Four parts. Take any one away and the line stops producing decisions.

Rank them by what you cannot rent

Before the answer: put the four in order, from the one you could have tomorrow to the one you could not.

Component	Can you rent it?	How long to have it
1 • Data pipeline	?	?
2 • Algorithm development	?	?
3 • Experimentation platform	?	?
4 • Software infrastructure	?	?

Takeaway Ranked one way you get a starting point. Ranked the other you get the bottleneck.

Rank them by what you cannot rent

Almost everyone builds 2 and 4 first, because they demo in a fortnight. The bottleneck is 1, and often 3.

Component	Can you rent it?	How long to have it
2 • Algorithm development	Yes — completely	An API key and an afternoon
4 • Software infrastructure	Yes — commodity	A quarter. Cloud, vector stores, orchestration
3 • Experimentation platform	Partly	Only if you have outcomes instrumented — most do not
1 • Data pipeline	No — not at all	Years. What makes it valuable is that it is yours

Diagnostic question for any AI proposal: where does the data come from every single day, and who owns that path?

Takeaway The sequence that produces demos and the sequence that produces value are different.

Data readiness — five tests, not one

“We have a lot of data” is not readiness. Read the stack bottom-up: raw history at the base, usable by a team at the top. Most organisations pass the bottom and fail near the top.

Accessible Can the people who would build on it actually get it, lawfully?

Governance, not technology

Consistent Does it mean the same thing across time and source?

Definitions changed and nobody wrote it down

Quality-controlled Has someone owned its correctness, continuously?

No owner between departments

Standardised One consistent format, one definition of the key entity?

Eleven systems, six formats, three definitions

Long Does the history go back far enough to contain the variation you care about?

Rarely the blocker

GraphCast exists because ERA5 exists — four decades, standardised, quality-controlled, public, free (Pre-read 4). Ask: what is our industry's ERA5, and if there is none, who would have to build it?

Takeaway A data-readiness failure is usually a governance finding, not a technical one.

Which of these four is the most data-ready?

A · Ten years of scanned supplier contracts, PDF

B · Three years of ticket resolutions, one helpdesk system, outcome logged

C · Every customer email since 2015, across four mailboxes

D · Fifteen years of sensor readings, format changed twice, undocumented

Volume is not readiness. Run the five tests on each before you answer.

SCALE — what happens at 10× volume?

Not a yes or no. Three shapes, and only one of them is worth hunting.

SUPERLINEAR ▲ the prize

Cost rises FASTER than volume. Ten times the loans is not ten times the underwriters — it is also their supervisors, their training, the errors that arrive with them, and the queue that forms while they work.

Example • 20 officers at 6 loans a day. At 10× you need ~200 people, a layer of supervisors, and a QA function nobody put in the business case.

LINEAR a real win

Cost tracks volume, one for one. Ten times the documents, ten times the translation bill. Automation still pays here — it just does not compound.

Example • translating product manuals. Real money saved. Defend it as efficiency, not as strategy.

ALREADY FLAT no scale story

Cost barely moves with volume, because the work is already software. There is nothing left to win on this axis, however good the model is.

Example • a nightly batch report on 10× the rows. Not a fail — move the argument to scope or learning.

Where superlinear cost hides: supervision • error rates • queueing • hiring lead time. Not one of them is the line item you were looking at.

Takeaway Superlinear is the prize — and the one rooms under-claim, because the cost sits in someone else's budget.

SCOPE — does the asset have a second customer?

Not “could we reuse it one day”. Whether a second use case can ride on the first one’s pipeline without rebuilding it.

PASSES • ANT

Payments data made lending possible. Lending data made insurance possible. Each new business launched cheaper than the last, on the same data layer and the same distribution.

Four businesses, one pipeline. That is the whole flywheel.

PASSES • RETINAL SCREENING

One fundus photograph, captured for diabetic eye screening, is also readable for glaucoma and — in research settings — for cardiovascular risk. Same capture, same pipeline, more questions answered.

The second customer was another clinic, not another company.

FAILS • THE ONE-OFF

A model built for a single regulatory return. Correct, useful, and there is not a line of it another team can use for anything else.

Worth doing. Not worth calling a platform.

The practical version: before you build, name the second team who would use this pipeline. If you cannot name them, you are buying scale only.

Takeaway Scope is a design question, not a technology one — and the quiet returns live here.

LEARNING — does the world label it for free?

The label has to arrive without anyone being asked, correct, and on a clock. Almost nothing passes this one.

PASSES • CREDIT & FRAUD

Default and chargeback arrive on their own, correctly, whether or not you ever built a model. Nobody is asked and nobody can forget.

The label IS the business event. That is why financial services industrialised this first.

PASSES • CHURN

The customer either renews or does not. The answer lands on the renewal date, for nothing, and it is not a matter of opinion.

Free, correct, and unavoidable — the three conditions.

FAILS • SUMMARISATION

“We will have the team rate the outputs” is a cost line that GROWS with usage. It can never run faster than the people doing the rating.

Paid labelling wearing a flywheel's clothes.

A use case that only buys scale is still a good use case. It just does not compound — so cost it as efficiency and defend it as efficiency.

Takeaway Scale • scope • learning — different tests, different evidence. Learning is the rarest and the most valuable.

Which of these produces a FREE label — one the world supplies without anyone being asked?

A • Marketing copy, approved by a manager before it goes out

B • Demand forecasting — next month's actual sales arrive anyway

C • Document summarisation, with weekly quality scores from the team

D • Meeting notes, corrected by whoever ran the meeting

Free means nobody is asked, nobody is paid, and it arrives whether you want it or not.

The free-label test

Does using the system produce a correct label, for free, that flows back in and changes behaviour? Not “does it collect data” — everything collects data.

PASS

Credit • fraud • recommendations • ad targeting. The world supplies the label without being asked — a default, a chargeback, a click or its absence. It arrives correct, on a clock, whether or not you built anything.

These compound. Every use makes the next one better, at no extra cost.

FAIL

Summarisation • drafting • most customer service. The outcome is an opinion, nobody records it, and the only way to get a label is to pay a human to produce one.

“We will have humans rate the outputs” is a cost line, not a flywheel — and this is most enterprise use cases.

Say this out loud to a board: an efficiency project defended as an efficiency project survives. One sold as a strategic moat does not.

Takeaway A use case that only buys scale is still good. It just does not compound.

So what is this use case actually buying?

Three tests, and they are not worth the same. Once a use case passes one, this is what you have bought.

	What it buys you	How long the edge lasts	How often it is real
SCALE	The same decision, cheaper, as volume rises	Shortest. Your competitor can rent the same model by next Tuesday	Common — and most use cases buy only this
SCOPE	The second use case is cheaper than the first, because it rides the same pipeline	Medium. The pipeline is not rentable. Rebuilding it is a budget and a decade	Uncommon, and usually a design choice rather than luck
LEARNING	It gets better by being used, at no extra cost	Longest. A rival must run it as long as you have, and time is the one input nobody can buy	Rare. Almost nothing passes the free-label test

A use case can pass more than one. Ant passed all three — which is exactly why it is the exhibit, and why Pre-read 2's 6% is the number to hold beside it.

Takeaway Most use cases buy scale. Price those as efficiency. Fight for the rare ones that compound.

The same use case, two audiences

The diagnostic and the sponsor framing are independent axes. Lead with the first; land on the second.

Diagnostic (what it buys)	Cost reduction	New revenue	New capability
Scale	The common case — automate the unit decision	Rare — volume alone rarely sells more	No
Scope	Second use case rides the first one's pipeline	The usual route — same asset, new product	Sometimes
Learning	Compounding accuracy lowers cost over time	Personalisation, pricing, risk	Where this lives

Pre-read 2, function split: 56% cite cost savings in software engineering and manufacturing; 67% cite revenue gains in marketing and sales.

Takeaway Same analysis. Two vocabularies. Never change the analysis to suit the audience.

Network effects — and the one that needs AI

Three different things share one phrase. Only the third depends on the model — and it only works if the label is free.

Type	What makes it stronger	Example	Needs AI?
Direct	More users on the same side	A payment network — Alipay is useful because merchants accept it	No
Indirect / cross-side	More users on the OTHER side	Marketplace: merchants and consumers	No
Data	Usage improves the PREDICTIONS for everyone	Fraud scoring, credit limits, recommendations	Yes

A real data network effect needs all four:

DATA

FREE

NOT SELF-REPORTED

OWNED

Takeaway Claimed data network effects are common. Ones that pass all four words are not.

Build vs buy — at portfolio level

Week 2 asked this per layer. At portfolio level it is one question asked four times, and the answer is a margin decision.

What you must own. The data pipeline, the consent and provenance behind it, the evaluation set, and the workflow the decision lands in. These are the things that take years and cannot be bought.

Rule of thumb: own what a competitor could not have by next Tuesday; rent the rest.

What you should rent. The model, the orchestration framework, the vector store, the compute. Renting is not weakness — it is what made the entry cost collapse, and your competitor is renting the same things.

The catch: renting turns a fixed cost into a variable one. We put a number on that after the break.

Live example, 10 Aug 2026: NVIDIA and six asset managers are assembling over US\$500bn to finance AI compute — the rented layer, financed like infrastructure.

Takeaway Own the compounding assets. Rent the commodities. Know which is which.

Where advantage lives when the model is rented

Run the cull: list what a well-funded competitor could have by next Tuesday. Whatever survives is the candidate list.

Survives the cull	Why it is hard to copy	Time to copy
Regulatory permission	Licences and approvals that took years and cannot be bought	Years — sometimes never
Proprietary data with a lawful basis	Not data you hold — data you have consent, provenance and permission for	Years
Distribution	Being where the decision already happens, at the moment it happens	Months to years
Workflow integration	Replacing you means retraining people and rewiring processes	Measured in disruption
<i>The model, the framework, the cloud, the prompt</i>	<i>Anyone can rent the same ones</i>	<i>By next Tuesday</i>

This is a use-case SELECTION checklist, not a post-hoc explanation — prefer the use case sitting on data you already hold, in a workflow you already own.

Takeaway If your only advantage is the model, you have a feature, not an advantage.

What a data pipeline actually is

FOR BUILDERS

The component you cannot rent, unpacked — because “we need a data pipeline” is where most AI budgets quietly go.

PRE-READ 1

The parts. Ingestion from source systems • contracts that say what a field means and who may change it • freshness guarantees • lineage so you can answer “where did this number come from” • consent and provenance tracking • monitoring for silent drift.

For builders: the hard part is not moving bytes. It is the contract — the agreement about meaning.

Why it takes years. None of the above is a technical problem. Data owners must agree, systems never designed to talk must talk, and departments that disagree about definitions must reconcile.

The tell: if your pilot ran on a one-off extract, you have not built any of this yet.

This is also why component 3 — the experimentation platform — usually fails: you cannot A/B test outcomes nobody instruments.

Takeaway The pipeline is an organisational agreement wearing an engineering costume.

Where the 2020 framework has aged

Pre-read 1 is the spine of this week and it was written before ChatGPT. Two of its four load-bearing claims have moved.

The 2020 claim	What changed by 2026	What to do with it
Marginal cost approaches zero, so scale pays forever	Classic software freezes the costly thinking into code once and replays it. A model re-derives the answer every request — so it is never paid off	AI-native margins 50–60% vs 80–90%. C2 puts a number on it
The data and the algorithms are the moat	The model layer is rented — anyone can have frontier capability for the price of a coffee. The data pipeline is not	Moat = data, distribution, workflow, permission. The data layer comes planned (DBS) or unplanned (Ant)
Bottlenecks are internal — hiring, coordination, specialisation	The binding constraint is increasingly external, and it appears BECAUSE you removed the internal ones. It need not be a regulator	C3. The framework counts only internal ones
Evidence: Amazon, Google, Ant, Ping An	Selected exemplars — chosen because they succeeded. Pre-read 2 supplies the missing frequency: about 6%	Good on mechanism, silent on how often it pays

It arrives three ways:

REGULATOR • Ant

PLATFORM • Apple → Meta

PUBLIC • face recognition, 2020

Takeaway Take the mechanisms. Re-check every number — two of these four have already moved.

One we killed

INSTEAD OF THE GUEST LECTURE

FROM PRACTICE

A use case that looked strong and did not survive contact with the data, the cost, or the customer.

The bet. 2019. Predict cell-site degradation before the site degrades, for a large US carrier. Random forests over about 5,000 features. Worth millions a year in failures caught early.

What we believed: that the hard part was the model. It was accurate. It was never the problem.

What happened. A year. Three of us, plus the compute bill. It worked in the sandbox exactly as designed — and it never ran anywhere else. We stopped it.

The question: before I show you the tells — what would you have checked first?

Takeaway Every one of these was visible before the money was spent.

One we killed — and the tell we missed

INSTEAD OF THE GUEST LECTURE

FROM PRACTICE

Five tells. Each one was visible early, and each one is something you already learned this week.

The tell	What it looked like at the time	What it actually meant
1 • The data was never going to arrive daily	A clean pilot on a hand-assembled extract	Nobody owned the path the data takes every day, with permission
2 • There was no free label	“We’ll have humans rate the outputs” in the plan	A cost line, not a flywheel — it could never improve by running
3 • Cost per call, never cost per successful task	A cost model with no failure term	Nobody could say what being wrong cost
4 • The workflow could not absorb it	One step got faster; the org chart did not change	Cycle time was set by handoffs and approvals nobody touched
5 • It worked — then something outside stopped it	A regulator, a partner, a platform policy, trust	The bottleneck appeared BECAUSE it succeeded

Takeaway You cannot avoid every bad bet. You can find out sooner, and cheaper.

One we killed — and the tell we missed

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And it PASSED tell 2 — degradation labels itself, the network records it, nobody has to be asked. The best test in this deck, cleared, and the project still died.

Takeaway Killing early is a success. Write the kill condition before the pilot starts.

INSTEAD OF THE GUEST LECTURE

The intake screen — one page

What evidence buys a pilot, what buys a budget line, and who is allowed to say no — before anyone builds.

Gate	What you must show	Who decides
1 • Worth doing	Which of scale / scope / learning it buys — with the test, not the claim	Sponsor + the function that owns the metric
2 • Possible	The five data tests, and which one you fail	Whoever owns the source systems
3 • Affordable	Cost per successful task, with the failure term in it	Finance — C2 gives you this number
4 • Absorbable	The decision it changes, and the workflow step it lands in	The operating owner, not the AI team
5 • Killable	The baseline captured now, and the written condition under which we stop	Agreed before the pilot starts

Assembled from published evidence (Pre-reads 1–4), not from one company's process — check it against how intake actually works where you are.

Takeaway A pilot needs gates 1, 2 and 5. A budget line needs all five.

Why agents stall — and what unblocks them

INSTEAD OF THE GUEST LECTURE

PRE-READ 2

62% are experimenting with agents. 23% are scaling anywhere. No single function is above 10% — and the reason is not capability.

What they report stops them	What is actually underneath (Week 4)	What actually unblocks it
Inaccuracy — 74%	Errors compound across steps: $0.95^{20} \approx 0.36$. The pilot never sees it, because a human was silently the error-correction layer	Fewer steps · recoverable failure · measure the run , not the step
Security and risk — 64%, the top barrier	The first write is the governance cliff. A pilot reads; a deployment writes. Nobody is attacking you and you still cannot show control	Confirmation on irreversible actions · capped autonomy · audit trail · a defined reversal path
Cybersecurity — 72%	The adversarial subset: injection through content the agent reads, poisoned tools, exfiltration through a tool it may call	Week 6. This half genuinely does have technical answers
Knowledge gaps — 60%	Brownfield. The constraints that govern real work are unwritten, so the agent cannot read them — Devin: 2/8 greenfield, 0/8 brownfield	Write the constraints down. That is a documentation project, not an AI one

In a pilot, when the agent is wrong, the answer to “who is accountable?” is “nobody, it is a pilot.” In a deployment, somebody signs. That sentence is the 10% ceiling.

Takeaway Every one of those fixes is an accountability mechanism wearing engineering clothes.

GROUP ACTIVITY • 4 MIN

Run the screen on one use case from your own organisation.

1 • Name it — and who does it today

2 • Which does it buy: scale, scope or learning?
Use the test

3 • Which of the five data tests do you fail?

4 • Name one use case that failed where you
work — and the tell

4 minutes in threes • we will take two • question 4 is the one to bring back

CAPSULE 1 • TAKEAWAY

Choose it before you build it.

What it buys • whether the data can feed it • whether the workflow can absorb it. After the break we do the part that decides most of it — what it actually costs.